

MD. MUTASIM BILLAH ABU NOMAN AKANDA

Senior Machine Learning Engineer • Document AI

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Professional Summary

ML engineer and researcher with **3+ years** shipping production systems: OCR and document AI (including low-resource languages), biomedical imaging experiments (RSNA MRI/CT on Kaggle), web-scale crawling, voice interfaces, and RAG. **2+ years** as undergraduate teaching assistant in CS/AI (labs, grading, project mentoring). Full stack from literature and experimentation to deployment (Python, PyTorch, LLM/VLM optimization and quantization, FastAPI, Docker). Peer-reviewed publications in **ACM** and **Springer**.

Experience

- **Apurba Technologies Ltd.** Dhaka, Bangladesh (on-site)
Senior Machine Learning Engineer Sep 2025 – present
 - **Borno OCR (ICT Ministry)**: Enterprise Bengali OCR, layout analysis, PDF/image to editable DOCX.
 - * Integrated **dots.ocr** for Bengali; **10,000+** documents/day.
 - * Qwen2.5-VL 4-bit quantization: **~4×** memory reduction, **~40%** lower infrastructure cost.
 - * Bbox alignment: **+25%** conversion accuracy, **~80%** fewer manual fixes.
 - **AI Interview Bot**: Real-time voice interview platform—LiveKit WebRTC, session-isolated RAG, automated evaluation.
 - * End-to-end voice pipeline (STT/LLM/TTS): **~sub-3s** average turn latency; **4-tier** TTS fallback for **~99%** synthesis uptime.
 - * Session-isolated RAG: **10+** concurrent interviews with independent context; **zero** cross-session data leakage.
 - * Automated evaluation replacing **~20 min** manual review per candidate; job-level analytics across **multiple roles**.
- **Pixalate Inc.** Remote (California, USA)
Data Scientist Mar 2025 – Aug 2025
 - **Focus**: Ad Intel Crawler—AI-assisted ad-network detection at web scale.
 - **Impact**:
 - * Detection pipeline: **73%** accuracy, **2M+** sites, **50K+** pages/day.
 - * Deterministic matchers where suitable: **\$50K+/year** compute savings.
 - * Crawler orchestration: **1M+** requests/day (async).
- **Green Pants Studio** Remote (Texas, USA)
AI Engineer Jun 2024 – May 2025
 - **Focus**: Vendidit—eCommerce scraping, valuation, and data mapping.
 - **Impact**:
 - * Scraper: **+40%** throughput, **100K+** products/day.
 - * OLS valuation: **92%** accuracy; **60%** fewer pricing errors.
 - * Dockerized services: **50%** faster deploys; **35%** reliability improvement.
- **Apurba Technologies Ltd.** Dhaka, Bangladesh (on-site)
Machine Learning Engineer Mar 2023 – May 2024
 - **Focus**: Bengali OCR (ICT Ministry)—layout, recognition, scene text.
 - **Impact**:
 - * YOLOv8 DLA: **85%+** on Bengali documents.
 - * 8-bit quantized DLA: **2×** faster inference, **50%** less memory.
 - * Co-first author on **ACM SE'24** Bengali DLA study; co-author on **Springer** (ICTCS) paper.

Teaching Experience

- **University of Liberal Arts Bangladesh (ULAB)** Dhaka, Bangladesh (on-site)
Teaching Assistant, Department of Computer Science and Engineering Feb 2021 – Jan 2023
 - **Courses**: Introduction to Programming, Object-Oriented Programming, Artificial Intelligence, Software Engineering.

- **Responsibilities:**

- * Led lab sections and tutorials (**30+** students per course); prepared lab materials and assessed coursework.
- * Held office hours; mentored students on programming and AI/ML course projects.

Leadership & Service

- **ULAB Computer Programming Club** Dhaka, Bangladesh
President 2022 – 2023
 - Organized **4** CSE-wide competitive programming contests as UCPC President (2022–2023).
 - Take Off Summer 2022: **40+** participants; lead problem setter on a **6**-problem contest (CSE department co-hosted).

Technical Skills

- **Languages & APIs:** Python (primary), TypeScript, SQL; REST; modular services
- **ML / AI:** PyTorch, TensorFlow, scikit-learn, Hugging Face, OpenCV; EAST/CRNN OCR; QAT; vLLM, Ollama, faster-whisper, Piper (TTS)
- **Medical imaging:** pydicom, DICOM preprocessing, EfficientNet3D, 3D/video CNNs (R3D-18), KerasCV
- **LLM / NLP:** LangChain, RAG, ChromaDB, GPT/Qwen workflows, semantic search
- **Backend & data:** FastAPI, WebSockets, LiveKit/WebRTC, SQLite, Redis, PostgreSQL, pgvector, SQLAlchemy, Pydantic
- **Automation:** Playwright, Crawl4AI, browser-use, BeautifulSoup, Scrapy, async pipelines
- **Frontend & DevOps:** Next.js, React, Gradio, Docker, Compose, GitHub Actions, pytest, Vitest
- **Analytics:** pandas, NumPy, Matplotlib, Seaborn, MLflow, W&B
- **Collaboration:** Git/GitHub, Jira, Notion, Slack, Agile

Education

- **University of Liberal Arts Bangladesh (ULAB)** Dhaka, Bangladesh
B.Sc. Computer Science and Engineering Summer 2019 – Fall 2022
CGPA: 3.96/4.00. **Summa Cum Laude** (top 1%; conferred May 2024). Vice Chancellor's Honors List; Dean's List Scholarship (3×).
Coursework: Artificial Intelligence, Digital Image Processing, Algorithms, Data Structures, Statistics and Probability, Software Engineering.

Research & Technical Projects

- Voice Healthcare Agent** GitHub
 - **FastAPI** API and **Next.js 14** /call UI: REST/WebSockets for STT/TTS and agent turns; optional **LiveKit** WebRTC worker reusing same Ollama, faster-whisper, and Piper-backed TTS.
 - **SQLite**-backed tool execution (e.g. appointments); session persistence and summaries aligned across REST, WebSocket, and LiveKit paths.
 - Optional **MuseTalk** lip-sync avatar pipeline; pytest (backend), Vitest (frontend), GitHub Actions CI.
 - *Python, TypeScript, faster-whisper, Piper, Ollama, Docker Compose (LiveKit).*

- SenseScan (Bengali handwritten OCR)** GitHub
 - EAST/LANMS detection, CRNN recognition (QAT-ready), reading-order assembly.
 - FastAPI: /plugin and /v1/ocr/handwritten; optional Gradio; DataParallel multi-GPU.
 - *PyTorch, FastAPI, OpenCV, LANMS, Pydantic.*

- GradConnectAI** GitHub
 - CV parsing, professor/opportunity discovery, evidence-based matching, outreach drafts.
 - *FastAPI, Next.js, TypeScript, PostgreSQL/pgvector, Playwright, Crawl4AI, Docker, GitHub Actions, Ollama/vLLM.*

RSNA biomedical imaging (Kaggle)

Kaggle

- Brain MRI: PyTorch EfficientNet3D on mpMRI DICOM (FLAIR, T1w, T1wCE, T2w) for MGMT promoter methylation classification.
- Abdominal CT: 2.5D EfficientNet CNN with 5-fold CV, 3D R3D-18 on DICOM slices, and DICOM-to-3D tensor preprocessing for RSNA trauma detection.
- *PyTorch, pydicom, KerasCV, EfficientNet3D, R3D-18.*

Publications

- **Akanda, M.B.A.N.*, Ahmed, M.*, Rabby, A.S.A., & Rahman, F.:** *Optimum Deep Learning Method for Document Layout Analysis in Low Resource Languages*. ACM Southeast Conference (ACM SE'24), 199–204. doi:10.1145/3603287.3651184. *Equal contribution.
- **Akanda, M.B.A.N., Prodhan, M., Sarwar, S., Raatul, A.M., Paul, B.:** *Voice Controlled Home Automation with Cloud-Based Environment Monitoring System*. ICTCS 2022, Lecture Notes in Networks and Systems, vol. 623, Springer, Singapore. doi:10.1007/978-981-19-9638-2_21

Honors & Awards

- **Problem setter:** ULAB Take Off Programming Contest (Summer 2022)
- **Vice Chancellor's Honors List:** Scholarship (Summer 2021)
- **2nd Runners-Up:** ULAB Take Off Programming Contest (Fall 2021)
- **Dean's List Scholarship:** Summer 2020, Fall 2020, Spring 2021
- **1st Runners-Up:** ULAB Take Off Programming Contest (Spring 2021)